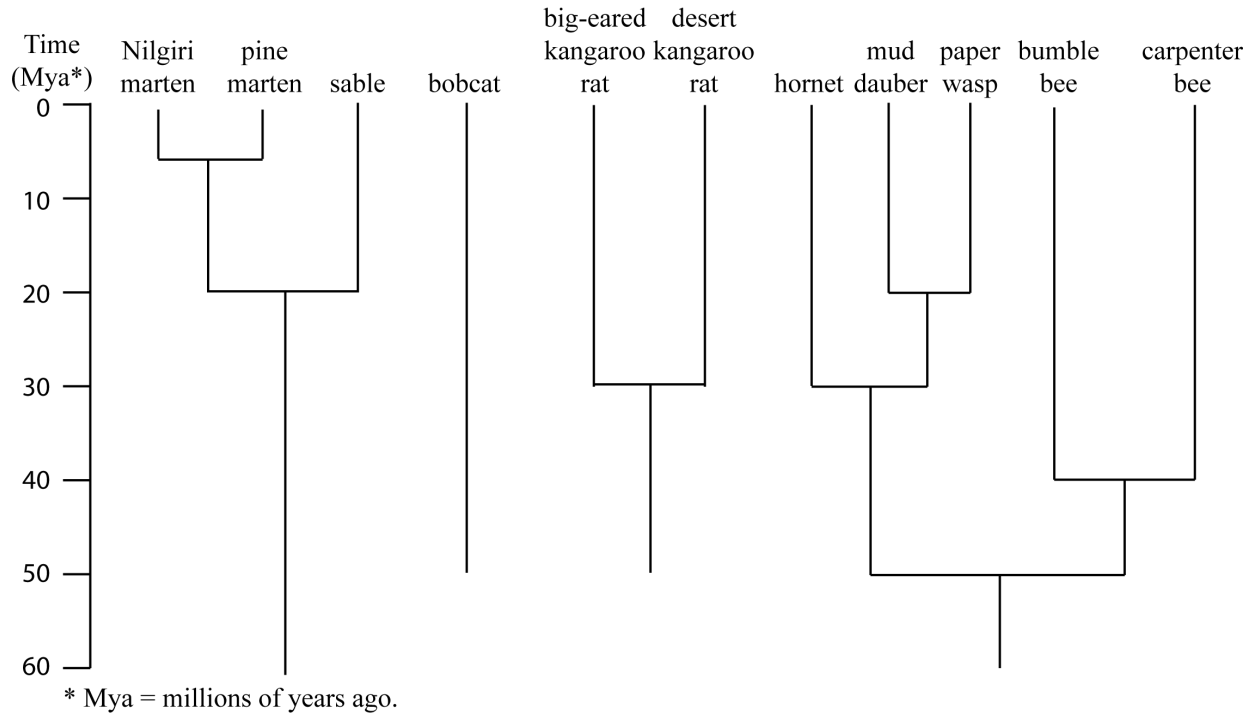


Read each question completely before answering. Write clear and concise answers.

- (4 pts) 1. Here is the full classification of the Scalloped Hammerhead Shark.
Eukaryota; Metazoa; Chordata; Chondrichthys; Carcharhiniformes; Sphyrnidae; Sphyrna; lewini
- (a) Which is the order?
- (b) Write the scientific name in proper format.
- (2 pts) 2. The type of scientific reasoning that uses a series of specific observations to develop general principles is called:
- (4 pts) 3. Clearly explain the difference between a belief and a scientific theory.
- (4 pts) 4. Write a clear definition of homology.

Consider the following hypothesis about the relationships among various organisms.



Answer questions 5–12 based on the hypothesis above. *Tell how you know for each answer!*

(3 pts) 5. Which two organisms shared a common ancestor *most recently*?

(3 pts) 6. Which are most closely related, Nilgiri marten and sable, or carpenter bee and bumble bee?

(3 pts) 7. Which organism(s), if any, is (are) most closely related to a hornet?

- (3 pts) 8. Approximately when did the common ancestor to big-eared kangaroo rat and desert kangaroo rat first appear?
- (3 pts) 9. Which organism(s), if any, is most closely related to the bobcat?
- (3 pts) 10. Suppose you find a homology between bumble bee and mud dauber. Is the hypothesis supported, falsified, or is the evidence inconclusive?
- (3 pts) 11. Suppose you find an analogy between desert kangaroo rat and bobcat. Is the hypothesis supported, falsified, or is the evidence inconclusive?
- (3 pts) 12. Suppose you find a homology between big-eared kangaroo rat and pine marten. Is the hypothesis supported, falsified, or is the evidence inconclusive?

- (7 pts) 13. List below the four missing steps of the Scientific Method. Then, for each step, write a statement that illustrates how scientists applied the scientific method to the experiments described below. I have provided step four as an example.

Houseflies are infamous for walking all over food around the house and at picnics. They stick out their proboscis (the fly version of a tongue) when they walk on food but not when they walk on other things, like the table. Scientists thought that flies might be able to taste the food with their feet, and stick out their proboscis whenever they identified something tasty. To test this idea, scientists filled one saucer with sugar water (sugar is a food source for the fly) and one saucer with plain water. They then glued live flies to the end of a slender stick. The flies were gently lowered until their feet just touched the liquid. If the liquid was plain water, the flies did not stick out their proboscis. If the liquid was sugar water, the flies always stuck out their proboscis.

(a)

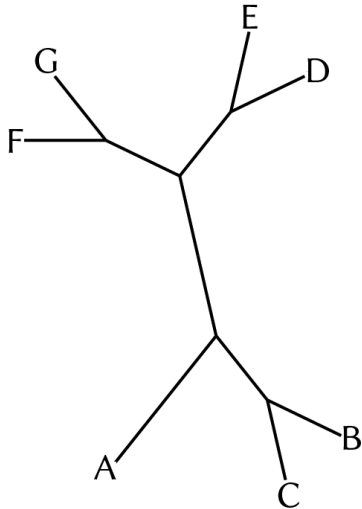
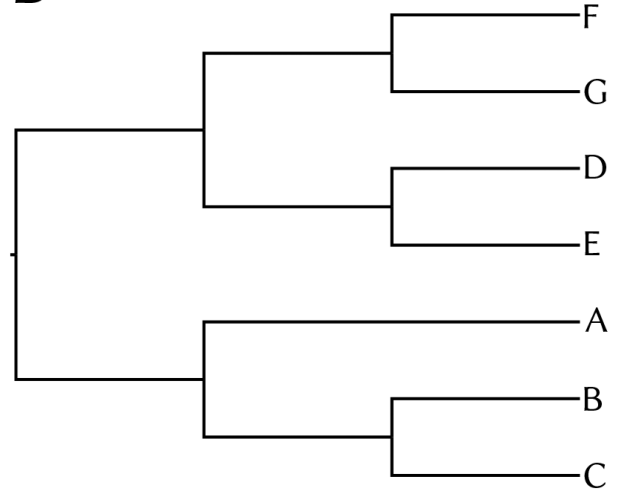
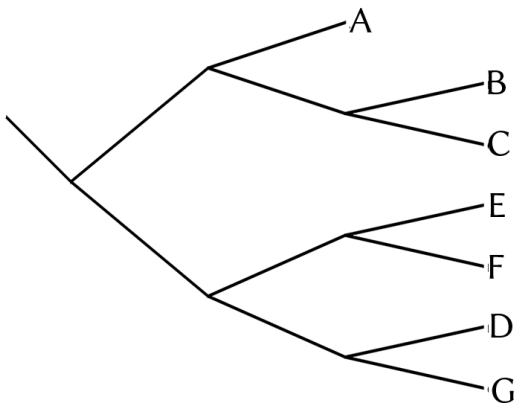
(b)

(c)

- (d) **Experiment and results:** Scientists noted that flies stuck out their proboscis when their feet touched sugar water but not when their feet touched plain water.

(e)

- (5 pts) 14. Below are several *correct* ways to draw phylogenetic trees. Which tree shows a different pattern of *relationships* compared to the others. Identify *all* differences between the tree you identified and the other three trees. Tell how you know why the relationships are different.

A**B****C****D**